Long Data

Emerging technology for improved access to and use of historic documentation
Emerging Technologies Taskforce · City of Boulder

Brian C. Keegan, Ph.D.
Associate Professor, Department of Information Science
University of Colorado Boulder
October 21, 2026

The council hotline

Council members send questions to staff through a shared email list, the council hotline. Staff answer on the same list, and the city publishes every message as open data

Since March 2023 the list has carried 779 messages, among them 207 posts of council questions and 142 staff answers

Many of those answers point to something the city had already written down: a budget table, a staff memo, an ordinance

The question for the next fifteen minutes

How many of council's questions could the city's own records answer, and could a machine find those records?

City of Boulder open data, "Emails to the City Council Hotline," March 2, 2023 to September 15, 2026

Long data

Big data means many records over a short span: clicks, sensor readings, card swipes

A city's records run the other way. One place, written down continuously for more than a century. Samuel Arbesman called this **long data**

Long data answers questions no snapshot can: what the code said in 1974, how a budget line moved over 26 years, whether an earlier council asked the same question

Most of it sits in a form a person can read and a computer can't

Holding	Readable?	Amount
Digital files with text	Yes	Small
Scanned pages	No	Large
Paper, film, audio	No	Unknown

Nobody has counted. The largest unread holding is probably audio: decades of council and board meetings, where the reasons behind decisions were spoken aloud

Samuel Arbesman, "Stop Hyping Big Data and Start Paying Attention to 'Long Data'," *Wired*, January 29, 2013

Document understanding

Document understanding is the research field, and the software, that turns a page into data a computer can use. Its main tasks:

- **Layout analysis** finds the headings, paragraphs, tables, and figures on a page, and the order to read them in
- **Optical character recognition (OCR)** reads printed or handwritten text from a picture of a page
- **Table recognition** rebuilds a table's rows, columns, and cells
- **Field extraction** pulls named items, such as a date or a total, out of a form

Tools for this work fall into three groups

- Converters that copy text already stored in a file: fast and exact, and blind to scans
- Chains of small models, one for each task, that run on an ordinary computer
- Vision-language models, which look at an image of the page and write the whole page out. The most capable, and the most expensive to run

None of them chooses which document to read, checks its own output, or decides what the public may see

Keegan, *Web Data Science*, Chapter 9, "Document Understanding" (github.com/cuinfoscience/Web-Data-Science-Book)

Data liberation

Data liberation moves information that is already public out of the formats that bury it (scans, PDFs, portals without an export button) and into data anyone can reuse, with a record of where every value came from

The records are public before and after. Liberation lowers the cost of using them

- **2009.** RECAP, a browser extension built at Princeton, copies federal court documents that users have already paid for into a free public archive
- **2014.** The Sunlight Foundation runs a PDF Liberation Hackathon to build shared tools for pulling tables out of government PDFs
- **2017 on.** The Public Utility Data Liberation project turns federal energy filings into one documented, open database
- **2020.** The U.S. Supreme Court rules in *Georgia v. Public.Resource.Org* that the state's annotated code cannot be copyrighted
- **Boulder.** Boulder Public Data rebuilds the county's precinct election results, 2004–2024, from published statements of vote

---

Tom Lee, "PDF Liberation: Why It Matters and How You Can Help," Sunlight Foundation, January 24, 2014; *Georgia v. Public.Resource.Org, Inc.*, 590 U.S. 255 (2020)

What liberation could do for the city

- **Hotline answers.** Staff drafting a reply to council find the memo that already answered it, with the page cited
- **Budget history.** A 26-year series in which every figure points to its page, rebuilt once instead of by every analyst who needs it
- **The law on a date.** Which version of an ordinance was in force when a permit was issued
- **Records requests.** A document made searchable once is ready for every later request that asks for it
- **Accessibility.** Every scanned page gains text that a screen reader can speak
- **Research and reporting.** Residents, reporters, and the university work from the same corpus as staff

Two liberation sprints

I tried this twice, on my own time, for a newspaper column

City budget books, 2002–2027

34 PDFs and 10,843 pages from the city's records portal. Six volumes, 1,651 pages, contain no text at all, only pictures of pages

Result: 450 figures across 26 years, each traced to its source document

State school yearbooks, 1986–1999

The text stored in the scans recovered 3.9% of one table's cells. OCR read the rest at 0.75 cents a page, about $5 for fourteen volumes

Result: of 1,269 figures read twice, 1,267 agree exactly

Reading the pages cost a few dollars. Finding the right pages, checking every figure, and recording every gap took nearly all of the effort

Source: Charting Boulder repository, budget-history and alternatives folders

Getting a report from the records portal

Chapter 9 of my textbook walks students through one folder of the city's records portal: the Finance Department's year-end revenue reports

The folder's page arrives empty, saying "Loading...". A script then asks the portal for the folder's contents. The chapter's code sends that same request

Every download gets checked. A real PDF starts with the characters `%PDF-`; for a scanned record, the portal sends back an empty file

```
\Central Records\Finance\Reports\Revenue Reports
7 entries
194346 2021 (folder)
194359 2022 (folder)
194372 2023 (folder)
194385 2024 (folder)
194398 Revenue Report 2018 (15 pages)
194399 Revenue Report 2019 (14 pages)
194400 Revenue Report 2020 (16 pages)

application/pdf 723920 b'%PDF-'
```

What the chapter's code printed in October 2026: the folder's contents, then the December 2024 report's file type, size, and first characters

One report at a time

The portal answers one request at a time. On October 6, 2026, it stopped answering after 13 to 17 requests in a row and came back after a five-minute pause

Reading the totals

Each report opens with a table of sales and use tax, the city's largest source of revenue, for this year and last

The code finds that table by its opening sentence rather than its page number, because the 2020 report put a restatement letter in front of it

It reads the totals row, then checks the row against the report's own arithmetic: this year minus last year must equal the change the report prints

```
('2017', '2018')
['$134,538,892', '$140,648,710',
 '$6,109,818', '4.54%']

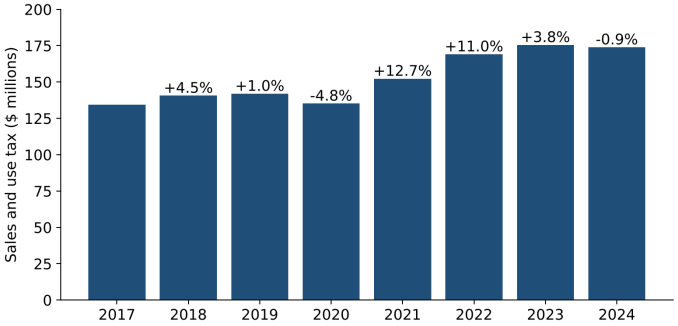
14 figures from 7 reports;
every change column checks
```

The 2018 report's two years, then its totals row: last year, this year, the change in dollars, and the change in percent

Two printings of 2018

Six years appear in two reports each. Five match to the dollar. The two printings of 2018 differ by $38,959, and neither report explains why

Eight years of sales and use tax



Down 4.76% in 2020, up more than 10% in 2021 and again in 2022, down 0.92% in 2024. Every bar traces to a page in a named report

Total sales and use tax, 2017–2024, read from seven year-end revenue reports; each year's latest printing; dollars of each year, not adjusted for inflation

Keegan, *Web Data Science*, Chapter 9, "Case Study: Boulder Revenue Reports"; the city's revenue reports, retrieved October 6, 2026

The hotline data

The city publishes the hotline under a public-domain license: 885 messages from March 2, 2023 to September 15, 2026

I set aside 106 messages sent from outside the list. Of the 779 that remain, 207 are council questions and 142 are staff answers

I kept each message's subject and the part its author wrote, dropping quoted replies, signatures, and boilerplate

Two sets of keyword lists then labeled every message: nine policy themes, and eight kinds of record an answer would need. One message can carry several labels

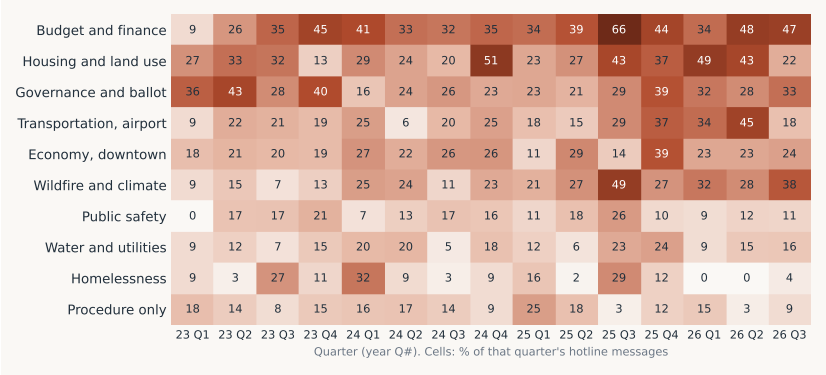
I read 25 messages by hand. The keywords named the main theme correctly in about 22

Limits

Keywords see words, not intent. And the attachments, where many answers live, are missing from the published data

Code, keyword lists, and labels: `code/hotline_provocation.py`, `code/hotline_themes.py`, `code/hotline-labels-all.csv`

What council asks about



Budget questions run through every quarter, from a third of messages in 2023 to 43% in 2025 and 2026. Wildfire and climate grew from 11% of messages in 2023 to 32% in 2026. Homelessness held near 13% for three years, then fell to 1% in 2026

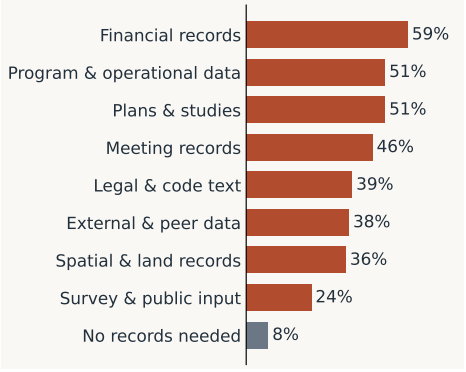
All 779 hotline-list messages, labeled by keyword; shares of each year's messages

What the answers needed

I sorted a random 40 of the 142 staff answers by what each one took. One coder, me, so treat the counts as a first look

The answer took	Answers
A resend, a logistics note, or a transmittal	10
A lookup: it already sat in a packet, memo, ordinance, or dataset	10
Facts from the record, plus judgment	14
New analysis or a policy position	6

Kind of record needed: share of all 349 questions and answers



A third were lookups

Of the 30 answers that did more than forward a file, 10 were lookups. Six needed judgment that no record holds. Financial records come up in 59% of questions and answers

What to liberate first

I ranked the city's holdings on three questions: how often council needs the record, how far back it runs, and how much personal information it carries

Start now

high demand, low privacy risk

- Budget books, revenue reports, audits
- Council packets and staff memos
- Ordinances, resolutions, and code versions

Next

high value, higher cost

- Minutes and meeting recordings
- Plans and consultant studies
- Board and commission appointments
- Program year-end reports

Only with screening

personal information

- Permits and land records
- Hotline mail and public comment

Accessibility

A scanned page is a picture. A screen reader, the software that reads a page aloud to a blind resident, finds nothing on it

Six volumes of the city's own budget books, 1,651 pages, are pictures of this kind

Federal rules under the Americans with Disabilities Act require the city's web content, posted PDFs included, to meet an accessibility standard (WCAG 2.1, level AA) by April 26, 2027. In April 2026 the Justice Department pushed that date back a year

Colorado's own law has applied since July 2024, and its grace period for good-faith effort ended in July 2025. A resident can now sue for $3,500 per violation

Owed in part

Documents residents use to get a service must be readable. Old scans kept in a marked archive, for reference only, are exempt, so making them readable is the city's choice

28 C.F.R. §§35.200–35.201, as amended by 91 Fed. Reg. 20902 (April 20, 2026); C.R.S. §24-34-802 (HB21-1110, HB24-1454)

The Colorado Open Records Act

Colorado declares that “all public records shall be open for inspection by any person at reasonable times,” except where the law says otherwise

A custodian has three working days to respond, or ten in extenuating circumstances. The legislature passed a longer deadline in 2025, and the governor vetoed it

A record kept in a searchable or sortable format must be released that way. A scan that is only a picture of a page can be released as a picture. The first hour of searching is free; after that the city may charge up to $41.37 an hour

Every request sends a person searching. A record liberated once is ready for the next request before anyone files it

Colorado Open Records Act, C.R.S. §§24-72-201, 24-72-203(3) and (3.5), 24-72-205(6); SB25-077 (vetoed April 17, 2025)

Community values

The city's Sustainability, Equity and Resilience Framework sets the goals behind its budget and comprehensive plan. Under **Responsibly Governed**, the city commits to a government that

- "Models stewardship, resilience, and sustainability of the city's financial, human, **information**, and physical assets"
- "Supports strategic decision-making with … timely and reliable data and analysis"
- "Enhances and facilitates **transparency**, accuracy, efficiency, effectiveness, and quality customer service in all city business"
- "Disaggregates data by race and ethnicity to ensure programs, policies, and financial decisions do not have a disparate impact on historically excluded community members"

The framework counts information among the city's assets, alongside money, people, and buildings. The records are the oldest part of that asset, and partners already help maintain it: the open data program works with local civic data groups, and the university law library has digitized the city charter

City of Boulder, "Sustainability, Equity and Resilience Framework," bouldercolorado.gov/sustainability-equity-and-resilience-framework

What happens next

Reading a scanned page costs under a cent today, and the price keeps falling

Finding the right document, checking what the machine read, and deciding what may be released will still take people. Retention schedules and legal review don't get cheaper with faster chips

Vendors will arrive with products running on the city's own documents. For the hotline, the product worth buying writes a draft answer with citations, which a staff member checks and signs



Buying a chatbot to answer council's questions

Publishing the records its answers would cite

Why I might be wrong

Demand may be low. My evidence is one coder and 40 hotline answers

Models may outgrow search. If a model can read every record at once, the city needs less search machinery, though it still needs every answer to cite a page

An automated assistant that chains several lookups can go wrong at each step without flagging it. Nobody has measured how often on city records

Most likely of all, the project just stops: multi-year records work dies on one budget cycle or one staff departure



Some records shouldn't become data

Sort every holding into one of three kinds: **already data**, **could become data**, or **shouldn't become data**

The third kind includes surveillance video, utility and recreation customer records, code complaints that name residents, protective-order addresses, and records about victims and juveniles

The danger is aggregation. Each record can stay obtainable on its own under CORA, while a searchable index would let anyone assemble a dossier in minutes

The city has done this before: the published hotline replaces every email address and phone number with a one-way code



"With great power comes great responsibility"

Who bears the cost

What gets scanned gets cited. Records left on paper drop out of every search and analysis, and the least-digitized records tend to describe the least powerful residents

OCR does worst on old, handwritten, tabular, and non-English pages, so search quality drops along the same lines

A searchable index exposes residents with the most contact with code enforcement and police far more than anyone else



Set the order on purpose

Let equity decide which holdings are digitized first. The city's own framework already asks it to check data for disparate impact

Key takeaways

1. The city holds **long data**, and most of it a computer can't yet read
2. Reading pages is cheap now; **finding and checking** them is the work
3. The **hotline** shows the demand: a third of substantive answers were lookups in records the city already holds
4. **Accessibility law, CORA, and the city's own framework** already call for this work
5. Before buying anything: **an inventory**, stable document IDs and bulk export written into contracts, and a rule that every machine answer cites a page

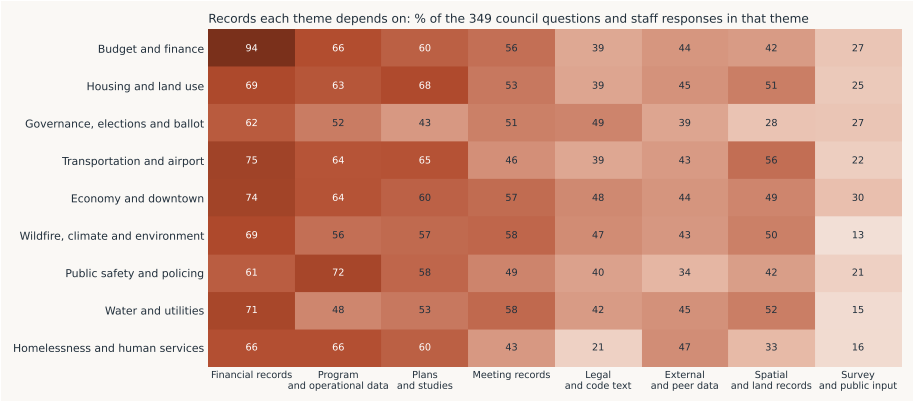
A test for the first demonstration

Which of the three kinds of holding is its corpus? Does every answer cite a document and page?

Appendix: record priorities in detail

Record	Serves	What we know of its state	Tier
Budget books	Financial questions; 26-year series	2002–2027 in the records portal; 1,651 pages lack text	Start
Revenue reports	Financial questions	In the records portal, 2018–2024; also sent as hotline attachments, which are not in the dataset	Start
Council packets and memos	Meeting records; plans; the answers themselves	Recent years in the agenda system; depth unknown	Start
Ordinances, resolutions, code versions	Legal text; which law applied when	Current code hosted outside the city; history in the records portal	Start
Minutes and recordings	Deliberation and intent	Unknown; needs inventory	Next
Plans and consultant studies	Plans and studies	Comprehensive plan: 15 editions converted, 1977–2026	Next
Board and commission appointments	Who served on what, when	Announced on the hotline; rosters unknown	Next
Program year-end reports	Operational data	Some sent as hotline attachments	Next
Permits and land records	Where things are	Department systems; custody split	Screen
Hotline and public comment	What council asks; deliberation	885 records, March 2023 to September 2026; email addresses and phone numbers coded	Screen

Appendix: which records each theme depends on



Financial records come up in more than 60% of questions and answers in every theme. Most answers need several kinds of record at once

Appendix: getting a report from the records portal

```
import requests, time
# PORTAL, LISTING, and DOWNLOAD hold the portal's addresses; chapter 9 gives them in full
HEADERS = {"User-Agent": "WebDataScience/1.0 (your-email@colorado.edu)"}
# The folder's listing: the request the folder page itself sends (194345 is Revenue Reports)
body = {"folderId": 194345, "getNewListing": True, "start": 0, "end": 100}
response = requests.post(PORTAL + LISTING, json=body, headers=HEADERS, timeout=60, verify=bundle)
for entry in response.json()["data"]["results"]:
    print(entry["entryId"], entry["name"])

time.sleep(30) # wait between requests to the city's server
response = requests.get(PORTAL + DOWNLOAD.format(docid=194397), headers=HEADERS, timeout=120, verify=bundle)
print(response.headers["Content-Type"], len(response.content), response.content[:5])
# application/pdf 723920 b'%PDF-'
```

`bundle` is a certificate file the book provides, because the city's server is set up in a way browsers forgive and Python doesn't. Chapter 9 also sends the request fields the page adds

Condensed from Keegan, *Web Data Science*, Chapter 9, "Getting the Files" (github.com/cuinfoscience/Web-Data-Science-Book)

Appendix: reading and checking the totals row

```
import pdfplumber

def row_words(page, label):
    """Return the words to the right of label, on the line where label is."""
    found = page.search(label)[0] # where the label sits on the page, in points
    line = page.crop((0, found["top"] - 1, page.width, found["bottom"] + 1))
    return [w["text"] for w in line.extract_words() if w["x0"] > found["x1"]]

for name in REPORTS:
    with pdfplumber.open(name) as pdf:
        page = find_summary_page(pdf) # the page whose text includes "This report provides"
        earlier, this, change = [to_number(t) for t in row_words(page, "Total Sales and Use Tax")[:3]]
    if this - earlier != change: # the report's own change column is the check
        print(f"{name}: {this:,} - {earlier:,} is not {change:,}")
# (nothing printed: every change column checks)
```

`to_number` turns "$134,538,892" or "(1,145,528)" into a whole number; accountants write a negative number in parentheses

Condensed from Keegan, *Web Data Science*, Chapter 9, "Reading the Totals Row by Position" (github.com/cuinfoscience/Web-Data-Science-Book)